

STORY AGENT · PILLAR

Infrastructure & Operations

How this pillar works and why it matters.

Team assembled by Riker: Jean-Luc Picard, Data, Geordi La Forge, Miles Edward O'Brien.

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Sovereign Infrastructure for Autonomous Agents

- Cost-Optimized Execution: Orchestrator directs work to cheapest capable model (Anthropic-free tier), with per-deliberation cost transparency (~\$0.002–0.003) and auto-remediating security (WorfGate).
- Stateful, Portable RAG: pgvector recall→act→store cycle compounds knowledge; retrieval is detached from vendor lock-in.
- Control Plane Integrity: Live async-status tracking, timeout guardrails, and shadow-testing ensure autonomy is measurable before deployment.

Why it matters: You get production-grade agent ops—cost-predictable, auditable, and incrementally autonomous—without rebuilding the wheel.

Infrastructure That Enforces Operational Discipline

- AWS Fargate deployment; WorfGate credential broker guarantees secrets are never logged, with auto-remediating green/yellow/red governance on every agent action
- Quark routing selects the cheapest adequate model per task — premium orchestration only at tier-4 Anthropic — yielding a measured \$0.002–\$0.003 per deliberation with full cost attribution in the control lane
- pgvector RAG pipeline (recall → act → store) runs portably across environments; async-status layer delivers live progress visibility and enforces timeout guardrails across all 150 MCP tools

Why it matters: You get a self-hosted, cost-attributed, credential-safe autonomous coding platform where every architectural constraint is mechanically enforced, not aspirationally documented.

Ship-Ready Infrastructure

- Zero-trust execution: WorfGate isolates credentials, auto-remediates failures (green/yellow/red), and enforces runtime guardrails without manual intervention.
- Cost-optimized scale: Quark routing dynamically selects the cheapest adequate model per task, with control-lane attribution (~\$0.002-0.003/deliberation).
- Stateful autonomy: RAG (pgvector) compounds knowledge across runs; async-status provides live progress + timeout failsafes for long operations.

Why it matters: Self-healing, cost-transparent infrastructure that scales with demand while keeping secrets and failures contained.

Miles Edward O'Brien's view

- Automated deployment to AWS Fargate for scalable and reliable execution
- Implemented WorfGate credential broker for secure secrets management and auto-remediating governor for fault tolerance
- Integrated async-status for live progress tracking and timeout guardrails to prevent resource exhaustion

Why it matters: By prioritizing infrastructure and operations, we ensure the Story Agent platform delivers consistent and reliable performance for our clients.

END OF PILLAR

Infrastructure & Operations — recap

- **Jean-Luc Picard:** Sovereign Infrastructure for Autonomous Agents
- **Data:** Infrastructure That Enforces Operational Discipline
- **Geordi La Forge:** Ship-Ready Infrastructure
- **Miles Edward O'Brien:** Miles Edward O'Brien's view

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